6 **(a)** **(i)**

$$P = \frac{V^2}{R}$$

$$R = \frac{V^2}{P} = \frac{(6.0)^2}{3.0} = 12\ \Omega$$

(ii)

$$R = \frac{\rho L}{A}$$

$$\rho = \frac{RA}{L}$$

$$= \frac{(12)\left(\pi \times \left(\frac{78 \times 10^{-6}}{2}\right)^2\right)}{0.020}$$

$$= 2.8670 \times 10^{-6} = 2.87 \times 10^{-6}\ \Omega\ \text{m}$$

(iii) The value of the resistivity calculated in (a)(ii) is about 50 times that stated in the table of constants.

The table of constants states the resistivity of tungsten at room temperature whereas the resistivity value in (a)(i) is the resistivity at a much higher temperature when the light bulb is in use.

When temperature increases, the lattice ions in tungsten gain thermal energy and vibrate with larger amplitudes. This increases the collisions between the free electrons and the lattice ions which hinder the movement of the electrons. Hence, resistivity increases with increasing temperature.

(b) **(i)**

p.d. across Y, $V_Y = \left(\dfrac{\frac{R}{2}}{\frac{R}{2} + \frac{R}{4}}\right)(6.0) = 4.0\ \text{V}$

current through Y, $I_Y = \dfrac{V_Y}{R} = \dfrac{4.0}{12}\ \text{A}$

$$Q = I_Y t$$

$$= \left(\frac{4.0}{12}\right)(2 \times 60)$$

$$= 40\ \text{C}$$

(ii) Consider $\underline{I = Anvq = Anev}$.

Since both filaments are identical, Ane is constant. Hence the current I through the filament is directly proportional to the mean drift velocity v of the electrons in the filament i.e. $I \propto v$.

The current through Y is twice the current through Z as the potential difference across Y is twice the potential difference across Z (OR the total current flowing through 2 bulbs and 4 bulbs in parallel are equal).

Therefore, the mean drift velocity of the electrons in Y is twice that of the electrons in Z.

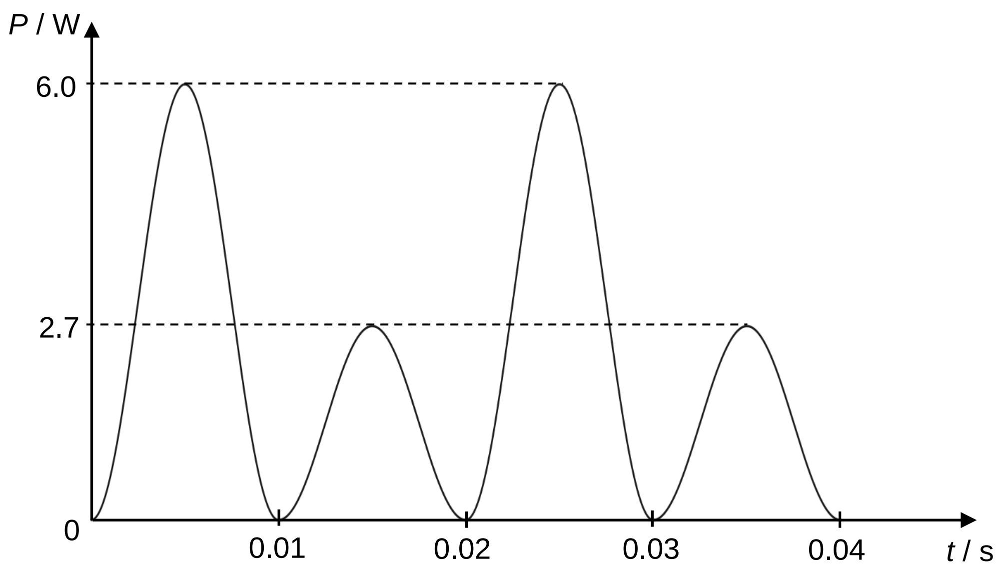
(c) **Diode in forward bias** *(current flows through the parallel arrangement of 2 bulbs on the left, and only through the diode on the right)* :

$V_Y = 6.0\ \text{V}$, peak power in $\text{Y} = 2(3.0) = 6.0\ \text{W}$

Diode in reverse bias *(current flows through the parallel arrangement of 4 bulbs on the right, and the parallel arrangement of 2 bulbs on the left)* :

$V_Y = 4.0\ \text{V}$, peak power in $\text{Y} = 2\left(\dfrac{4.0^2}{12}\right) = 2.6667 = 2.7\ \text{W}$

$T = \dfrac{1}{f} = \dfrac{1}{50} = 0.02\ \text{s}$



7 (a) (i) Solution 1